

Multivariate Procedures with R

Principle Component Analysis

:::

Principle Component and Its Graphical Analysis in R

Shalabh

Department of Mathematics and Statistics

Indian Institute of Technology Kanpur

Principal Component Analysis:

Let Σ be the a $(p \times p)$ symmetric and positive definite covariance matrix of $(p \times 1)$ random vector $\underline{X} = (X_1, X_2, \dots, X_p)'$.

Let the eigenvalues (characteristic roots) of Σ be $\lambda_1, \lambda_2, \dots, \lambda_p$ such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. Let the eigen vectors (characteristic vectors) of $\lambda_1, \lambda_2, \dots, \lambda_p$ be $\underline{e}_1, \underline{e}_2, \dots, \underline{e}_p$.

The i^{th} principle component is given by linear combination

$$Y_i = \underline{e}_i' \underline{X} = e_1 X_1 + e_2 X_2 + \dots + e_p X_p, i = 1, 2, \dots, p$$

with $Var(Y_i) = \underline{e}_i' \Sigma \underline{e}_i = \lambda_i, i = 1, 2, \dots, p$

and $Cov(Y_i, Y_k) = \underline{e}_i' \Sigma \underline{e}_k = 0, i \neq k$.

If some λ_i 's are equal, the choices of the $\underline{e}_i$'s, and hence Y_i , are not unique.

Principal Component Analysis: Proportion of Variance

The proportion of total variance accounted by

- First principal component is $\frac{\lambda_1}{\lambda_1 + \lambda_2 + \dots + \lambda_p}$
- Second principal component is $\frac{\lambda_2}{\lambda_1 + \lambda_2 + \dots + \lambda_p}$
-
- p^{th} principal component is $\frac{\lambda_p}{\lambda_1 + \lambda_2 + \dots + \lambda_p}$

Principal component technique helps in dimensionality reduction based on these proportion of variances.

Principal Component Analysis in R:

Description

Performs a principal components analysis on the given data matrix and returns the results as an object of class `prcomp`.

Usage `prcomp(x, ...)`

```
prcomp(formula, data = NULL, subset,...)
```

```
prcomp(x, retx = TRUE, center = TRUE, scale. =  
FALSE, tol = NULL, rank. = NULL, ...)
```

Arguments

formula a formula with no response variable, referring only to numeric variables.

data an optional data frame containing the variables in **formula**.

x a numeric or complex matrix (or data frame) which provides the data for the principal components analysis.

Principal Component Analysis in R:

Arguments

retx a logical value indicating whether the rotated variables should be returned.

center a logical value indicating whether the variables should be shifted to be zero centered.

scale. a logical value indicating whether the variables should be scaled to have unit variance before the analysis takes place.

tol a value indicating the magnitude below which components should be omitted.

rank. optionally, a number specifying the maximal rank, i.e., the maximal number of principal components to be used.

newdata An optional data frame or matrix in which to look for variables with which to predict. If omitted, the scores are used.

Principal Component Analysis in R:

Value

prcomp returns a list containing the following components:

sdev the standard deviations of the principal components.

rotation the matrix of variable loadings (i.e., a matrix whose columns contain the eigenvectors).

x if **retx** is true the value of the rotated data (the centred (and scaled if requested) data multiplied by the rotation matrix) is returned.

center, scale the centering and scaling used, or **FALSE**.

Principal Component Analysis in R: Example

Let $\Sigma = \begin{pmatrix} 11 & -12 & 6 \\ -12 & 15 & 4 \\ 6 & 4 & 12 \end{pmatrix}$ be the a (3×3) symmetric and positive definite covariance matrix of random vector $\underline{X} = (X_1, X_2, X_3)'$.

`x1 = c(11, -12, 6)`

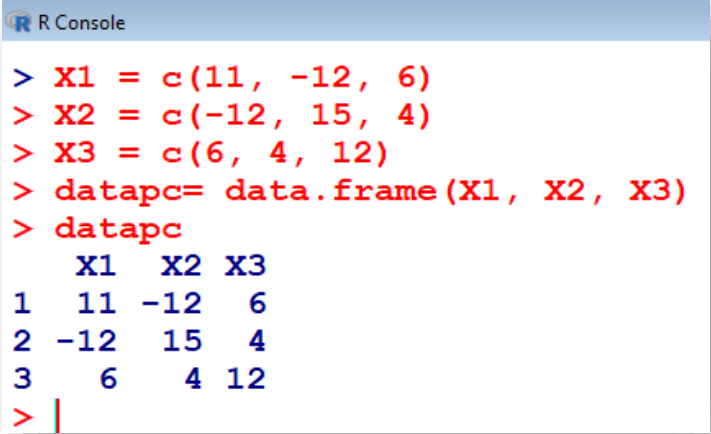
`x2 = c(-12, 15, 4)`

`x3 = c(6, 4, 12)`

`datapc= data.frame(x1, x2, x3)`

`datapc`

	x1	x2	x3
1	11	-12	6
2	-12	15	4
3	6	4	12



```
R Console
> x1 = c(11, -12, 6)
> x2 = c(-12, 15, 4)
> x3 = c(6, 4, 12)
> datapc= data.frame(x1, x2, x3)
> datapc
  x1 x2 x3
1 11 -12  6
2 -12 15  4
3  6  4 12
> |
```

Principal Component Analysis in R: Raw Data

```
pcdataout = prcomp(datapc)
```

```
pcdataout
```

```
Standard deviations (1, ..., p=3):
```

```
[1] 1.783783e+01 5.460014e+00 3.787415e-16
```

```
Rotation (n x k) = (3 x 3):
```

	PC1	PC2	PC3
x1	-0.6619978	0.4807521	0.5750098
x2	0.7454701	0.5018419	0.4386673
x3	-0.0776738	0.7190494	-0.6906046

The principal components are

$$Y_1 = \underline{e}_1' \underline{X} = -0.66X_1 + 0.74X_2 - 0.07X_3 \text{ with } Var(Y_1) = 17.8^2,$$

$$Y_2 = \underline{e}_2' \underline{X} = 0.48X_1 + 0.50X_2 + 0.72X_3 \text{ with } Var(Y_2) = 5.46^2,$$

$$Y_3 = \underline{e}_3' \underline{X} = 0.58X_1 + 0.44X_2 - 0.69X_3 \text{ with } Var(Y_3) = (3.78 \times 10^{-16})^2$$

Principal Component Analysis in R: Raw Data

```
> summary(pcdatout)
```

Importance of components:

	PC1	PC2	PC3
Standard deviation	17.8378	5.46001	2.272e-15
Proportion of Variance	0.9143	0.08567	0.000e+00
Cumulative Proportion	0.9143	1.00000	1.000e+00

Principal Component Analysis in R: Raw Data

R Console

```
> X1 = c(11, -12, 6)
> X2 = c(-12, 15, 4)
> X3 = c(6, 4, 12)
> datapc= data.frame(X1, X2, X3)
> pcdataout = prcomp(datapc)
> pcdataout
Standard deviations (1, ..., p=3):
[1] 1.783783e+01 5.460014e+00 2.272449e-15

Rotation (n x k) = (3 x 3):
      PC1      PC2      PC3
X1 -0.6619978 0.4807521 0.5750098
X2  0.7454701 0.5018419 0.4386673
X3 -0.0776738 0.7190494 -0.6906046
> summary(pcdataout)
Importance of components:
              PC1      PC2      PC3
Standard deviation    17.8378  5.46001 2.272e-15
Proportion of Variance  0.9143 0.08567 0.000e+00
Cumulative Proportion  0.9143 1.00000 1.000e+00
> |
```

Principal Component Analysis: Biplot

Description : Plot a biplot on the current graphics device.

Biplot helps in visualizing the similarities and dissimilarities between the samples, and shows the impact of each attribute on each of the principal components.

Usage `biplot(x, ...)`

```
biplot(x, y, var.axes = TRUE, col, cex =  
rep(par("cex"), 2), xlab = NULL, ylab = NULL,  
xlim = NULL, ylim = NULL, arrow.len = 0.1, main  
= NULL, xlab = NULL, ylab = NULL, ...)
```

Arguments

x The biplot, a fitted object.

y The second set of points (a two-column matrix), usually associated with variables.

Principal Component Analysis: Biplot

Arguments

var.axes If **TRUE** the second set of points have arrows representing them as (unscaled) axes.

col A vector of length 2 giving the colours for the first and second set of points respectively (and the corresponding axes).

cex The character expansion factor used for labelling the points.

arrow.len The length of the arrow heads on the axes plotted in **var.axes** is true. The arrow head can be suppressed by **arrow.len = 0**.

xlim, ylim Limits for the x and y axes in the units of the first set of variables.

Principal Component Analysis: Biplot

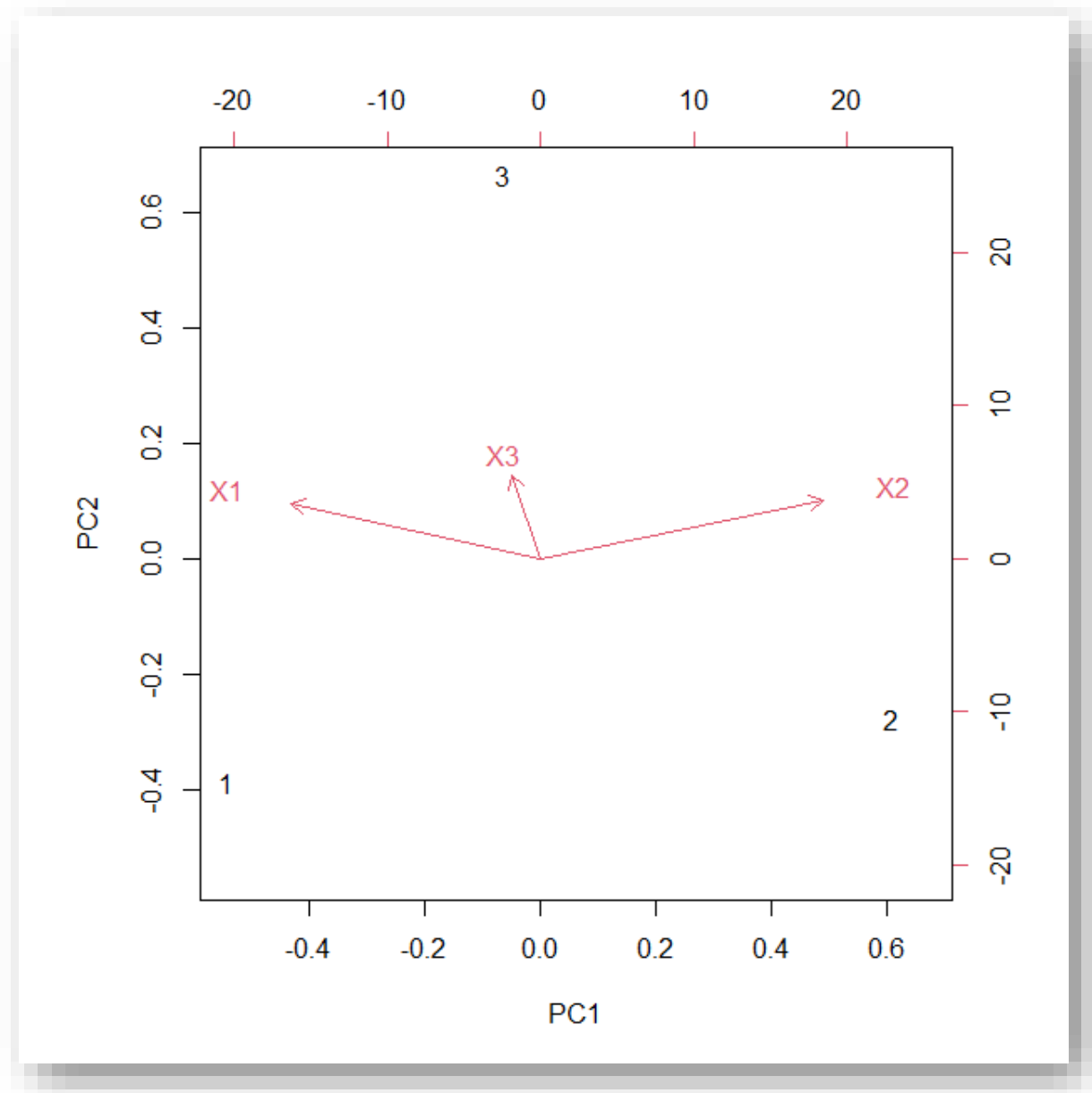
- The biplot of the principal component shows both PC scores of samples (dots) and loadings of variables (vectors).
- The further away these vectors are from a PC origin, the more influence they have on that PC.
- Representation of angle degrees between two variables :
 - 90° indicates no correlation between the two variables,
 - less than 90° indicates a positive correlation between two variables and
 - more than 90° indicates a negative correlation between two variable.

Principal Component Analysis: Biplot

- The vector line length represents the variance level of the variable:
 - the longer the vector line, the greater the variance.
 - the shorter the line, the less the variance.

Principal Component Analysis: Biplot of Raw Data

`biplot(pcdatout)`



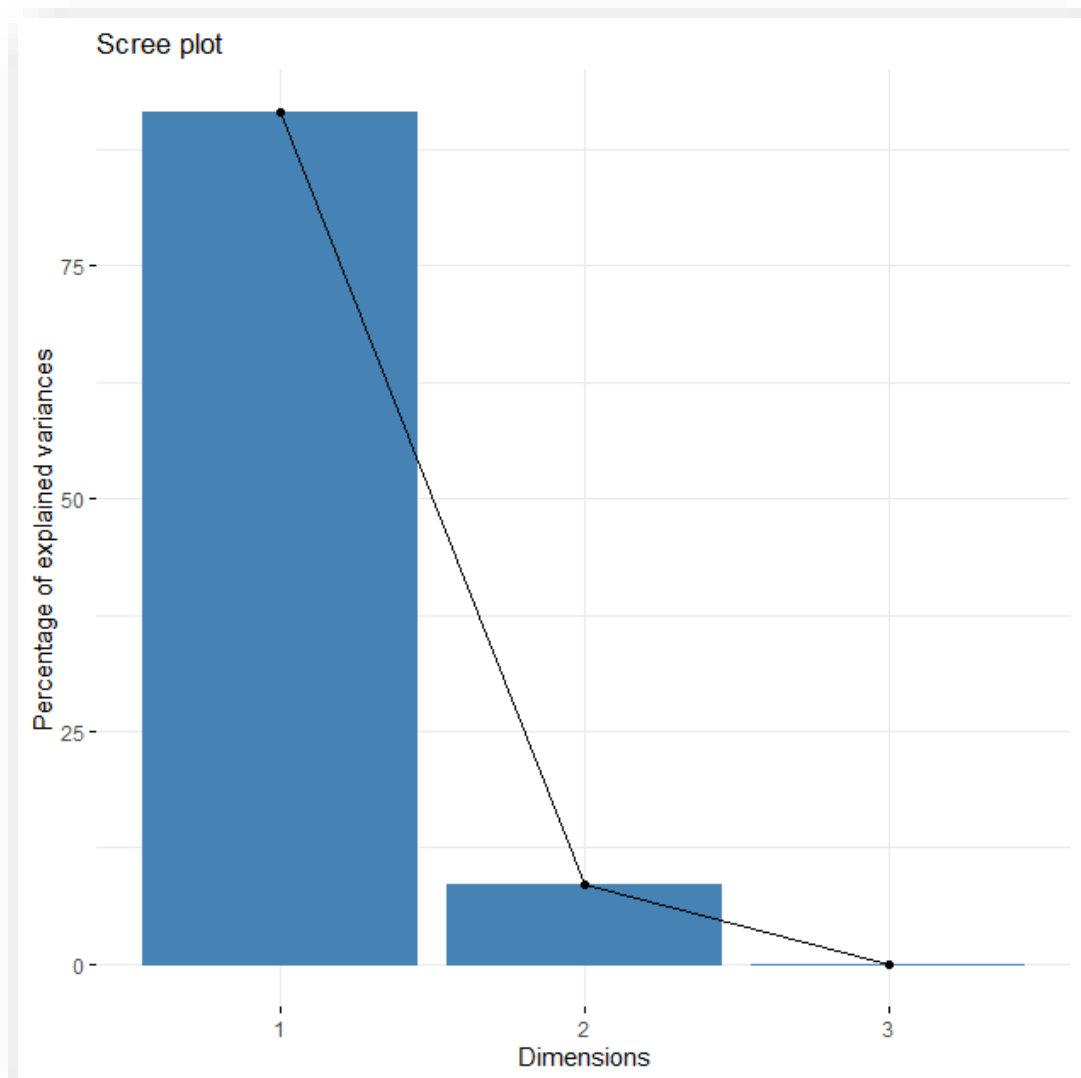
Principal Component Analysis: Scree Plot

- Scree Plot is used to visualize the importance of each principal component. A scree plot is a diagnostic tool to check whether PCA works well on the data or not.
- A scree plot shows how much variation each principal component captures from the data. It is used to determine the number of principal components to be retained.
- The scree plot has eigenvalues on y-axis and the number of principal components on x-axis.
- An ideal curve should be steep, then bends at an “elbow” — this is your cutting-off point — and after that flattens out.
- It is generated with the function `fviz_eig()`

Principal Component Analysis: Scree Plot

```
library(factoextra)
```

```
fviz_eig(pcdatout)
```



Principal Component Analysis: Contribution of Each Variable

Cos2 visualization is used to determine how much each variable is represented in a given component. It is computed using the function

`fviz_cos2`

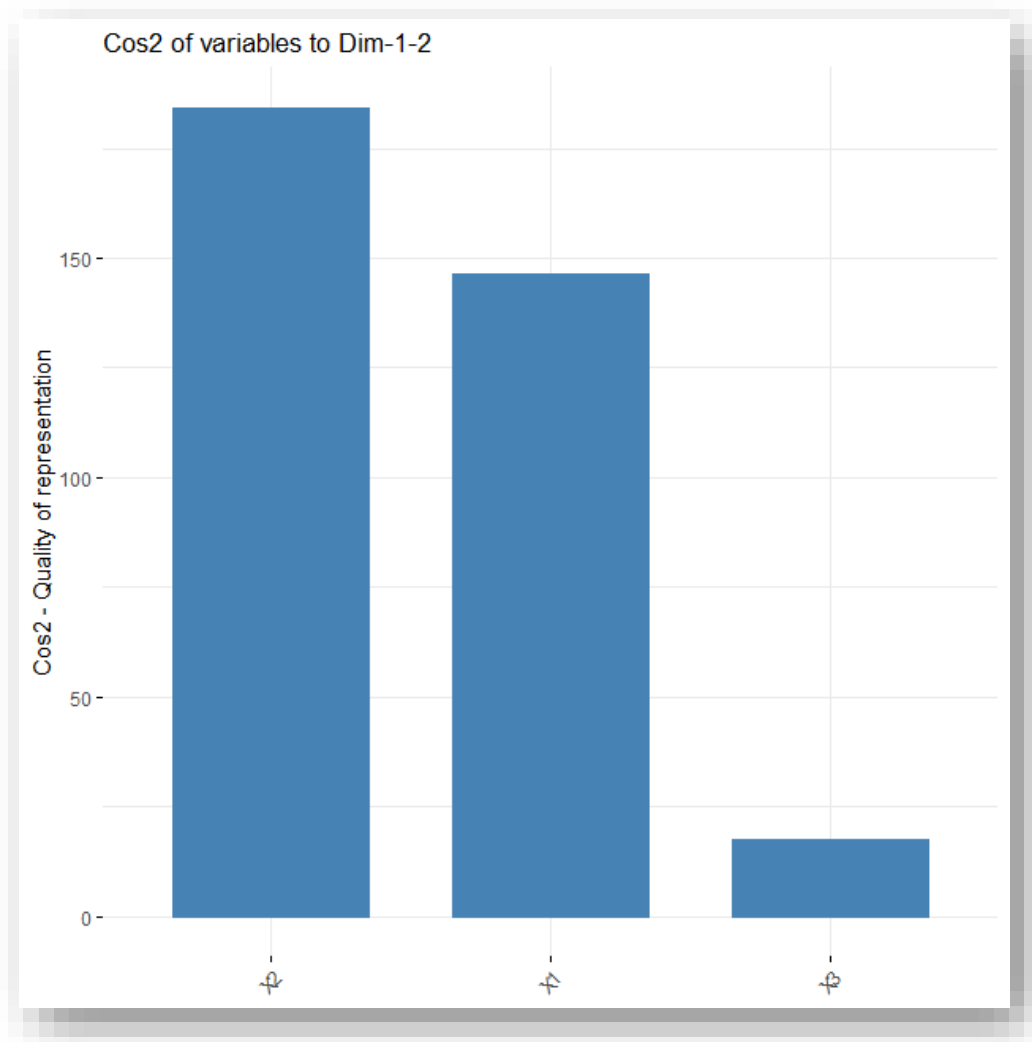
A high value indicates a good representation of the variable on that component.

A low value indicates that that component does not perfectly represent the variable.

Principal Component Analysis: Contribution of Each Variable

`library(factoextra)`

`fviz_cos2(pcdtaout, choice = "var", axes = 1:2)`



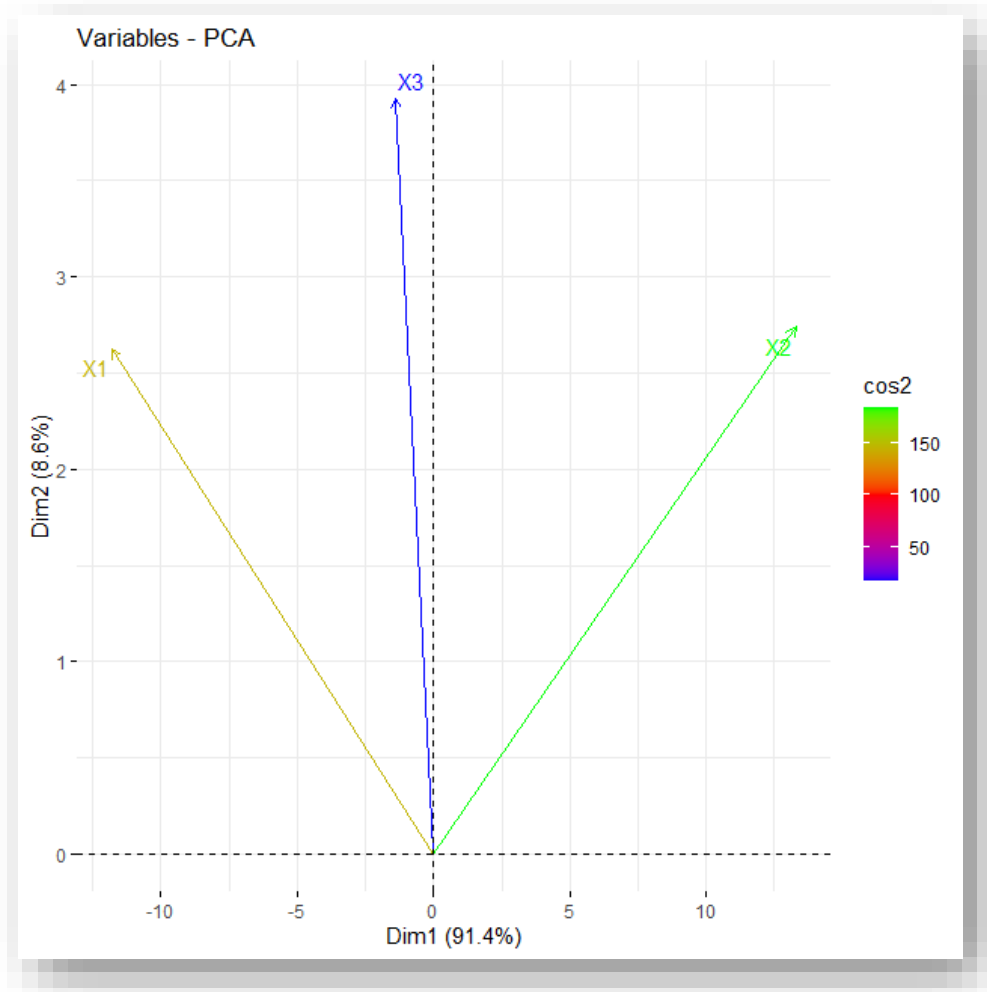
Principal Component Analysis: Biplot combined with cos2

Biplot and attribute importance can be combined to create a single biplot, where attributes with similar cos2 scores will have similar colours.

This is achieved by the `fviz_pca_var` function.

Principal Component Analysis: Biplot combined with cos2

```
fviz_pca_var(pcdatout, col.var = "cos2",  
gradient.cols = c("blue", "red", "green"),  
repel = TRUE)
```



Principal Component Analysis: Scaled data

```
pcdataoutcs = prcomp(datapc, center = TRUE, scale. =  
TRUE)
```

```
> pcdataoutcs
```

```
Standard deviations (1, ..., p=3):
```

```
[1] 1.455088e+00 9.395320e-01 1.316260e-16
```

```
Rotation (n x k) = (3 x 3):
```

	PC1	PC2	PC3
X1	-0.6859507	-0.06526322	-0.7247153
X2	0.6085311	0.49461231	-0.6205228
X3	-0.3989504	0.86665990	0.2995650

The principal components are

$$Y_1 = \underline{e}_1' \underline{X} = -0.69X_1 + 0.61X_2 - 0.39X_3 \text{ with } Var(Y_1) = 1.45^2,$$

$$Y_2 = \underline{e}_2' \underline{X} = -0.07X_1 + 0.49X_2 + 0.86X_3 \text{ with } Var(Y_2) = 0.939^2,$$

$$Y_3 = \underline{e}_3' \underline{X} = -0.72X_1 - 0.62X_2 + 0.29X_3 \text{ with } Var(Y_3) = (1.31 \times 10^{-16})^2$$

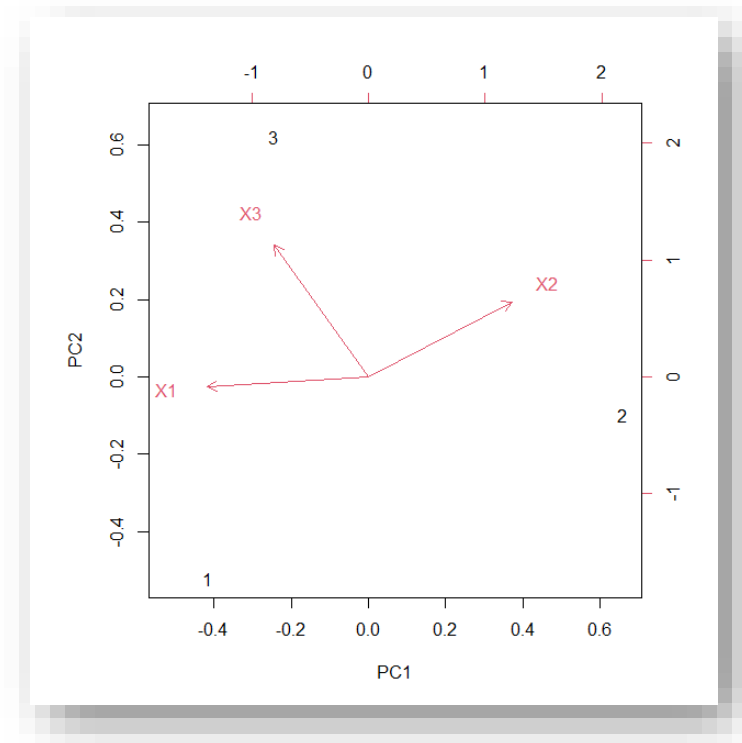
Principal Component Analysis: Scaled data

```
> summary(pcdatouts)
```

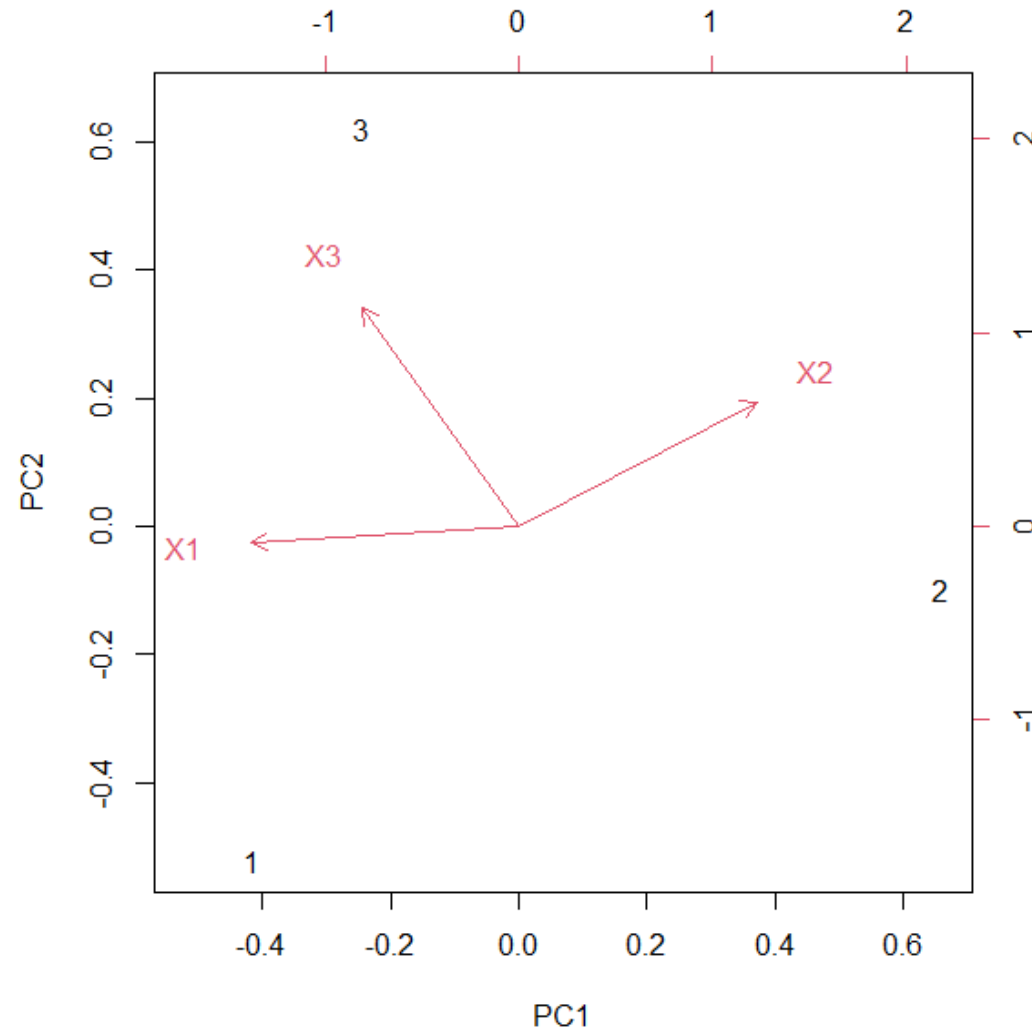
Importance of components:

	PC1	PC2	PC3
Standard deviation	17.8378	5.46001	2.272e-15
Proportion of Variance	0.9143	0.08567	0.000e+00
Cumulative Proportion	0.9143	1.00000	1.000e+00

```
> biplot(pcdatouts)
```



Principal Component Analysis: Biplot with Scaled Data



Principal Component Analysis: Scaled Data

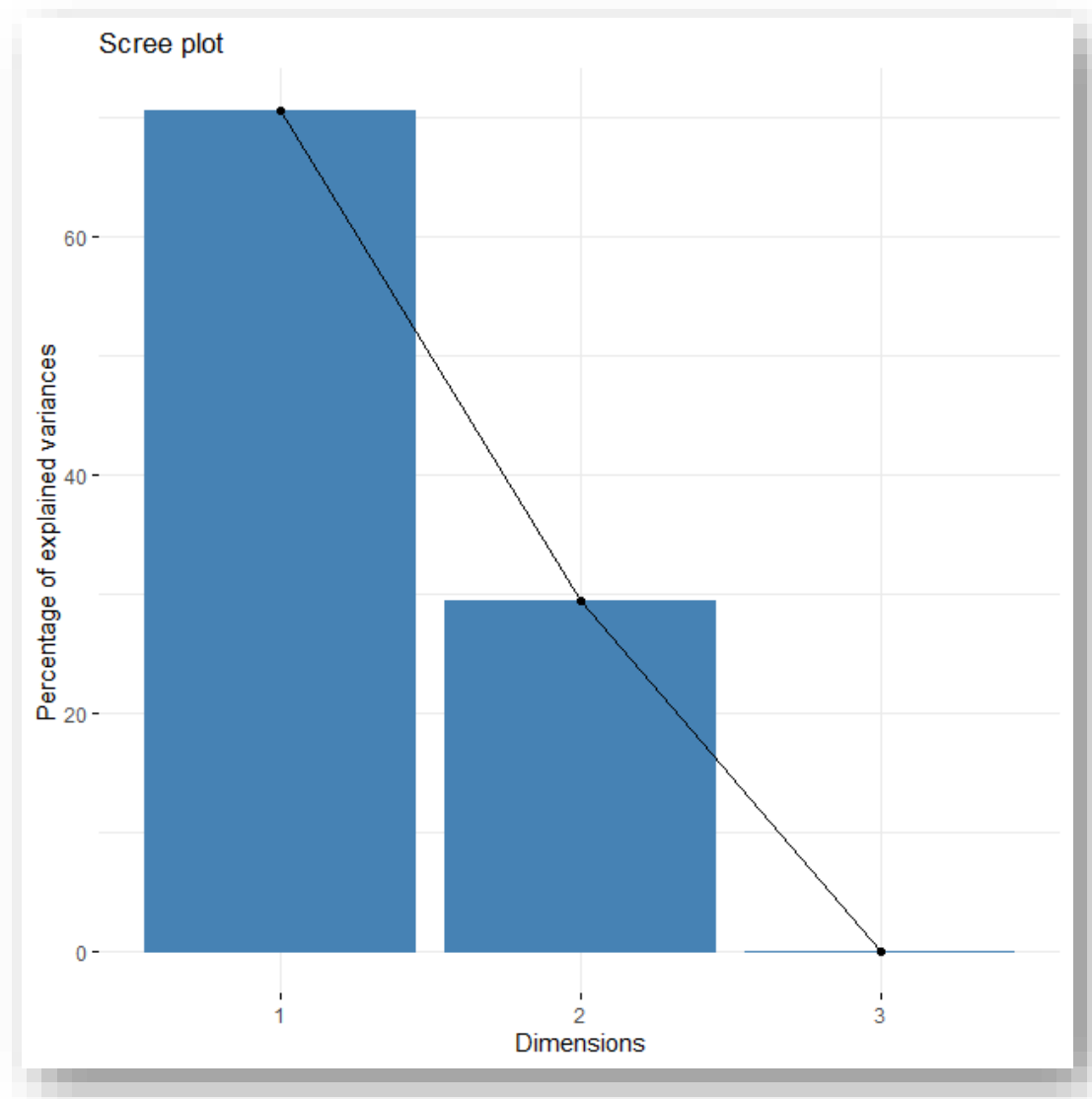
```
R Console

> X1 = c(11, -12, 6)
> X2 = c(-12, 15, 4)
> X3 = c(6, 4, 12)
> datapc= data.frame(X1, X2, X3)
> pcdataoutcs = prcomp(datapc, center = TRUE, scale. = TRUE)
> pcdataoutcs
Standard deviations (1, ..., p=3):
[1] 1.455088e+00 9.395320e-01 1.233994e-16

Rotation (n x k) = (3 x 3):
      PC1      PC2      PC3
X1 -0.6859507 -0.06526322  0.7247153
X2  0.6085311  0.49461231  0.6205228
X3 -0.3989504  0.86665990 -0.2995650
> summary(pcdataoutcs)
Importance of components:
      PC1      PC2      PC3
Standard deviation    1.4551 0.9395 1.234e-16
Proportion of Variance 0.7058 0.2942 0.000e+00
Cumulative Proportion 0.7058 1.0000 1.000e+00
> biplot(pcdataoutcs)
> |
```

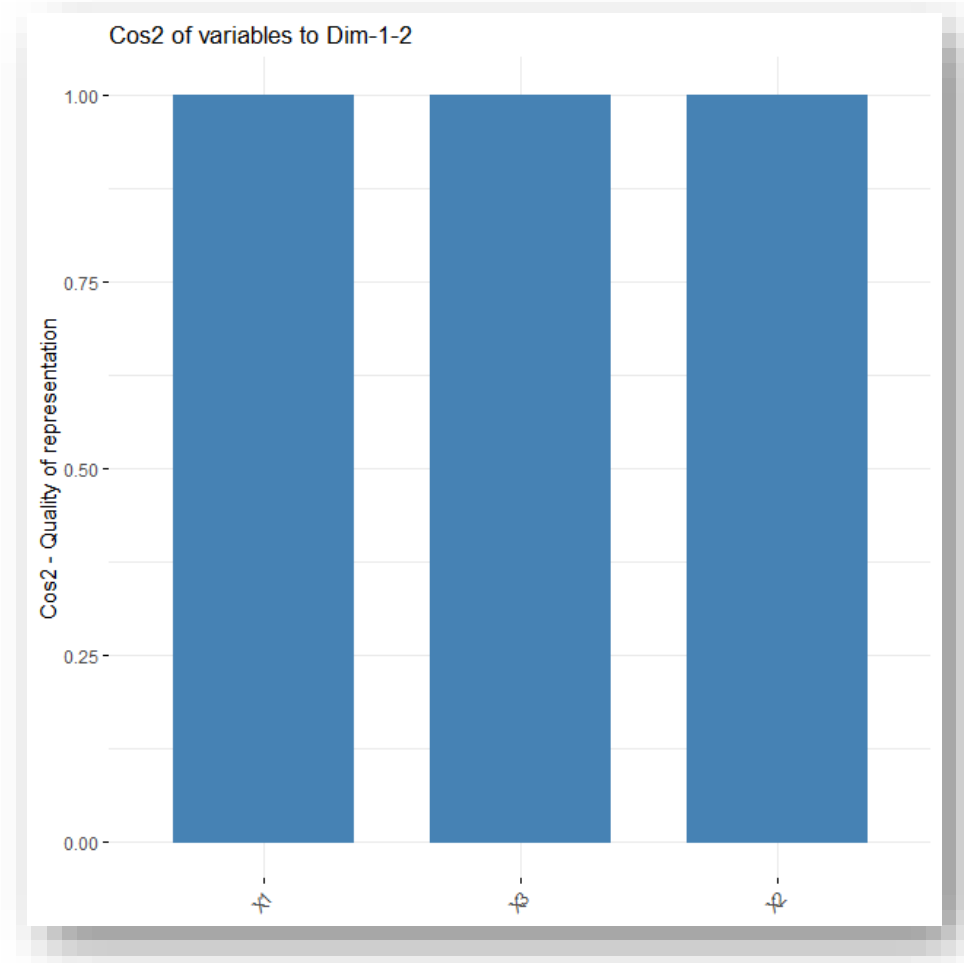
Principal Component Analysis: Scree plot of Scaled Data

`fviz_eig(pdataoutcs)`

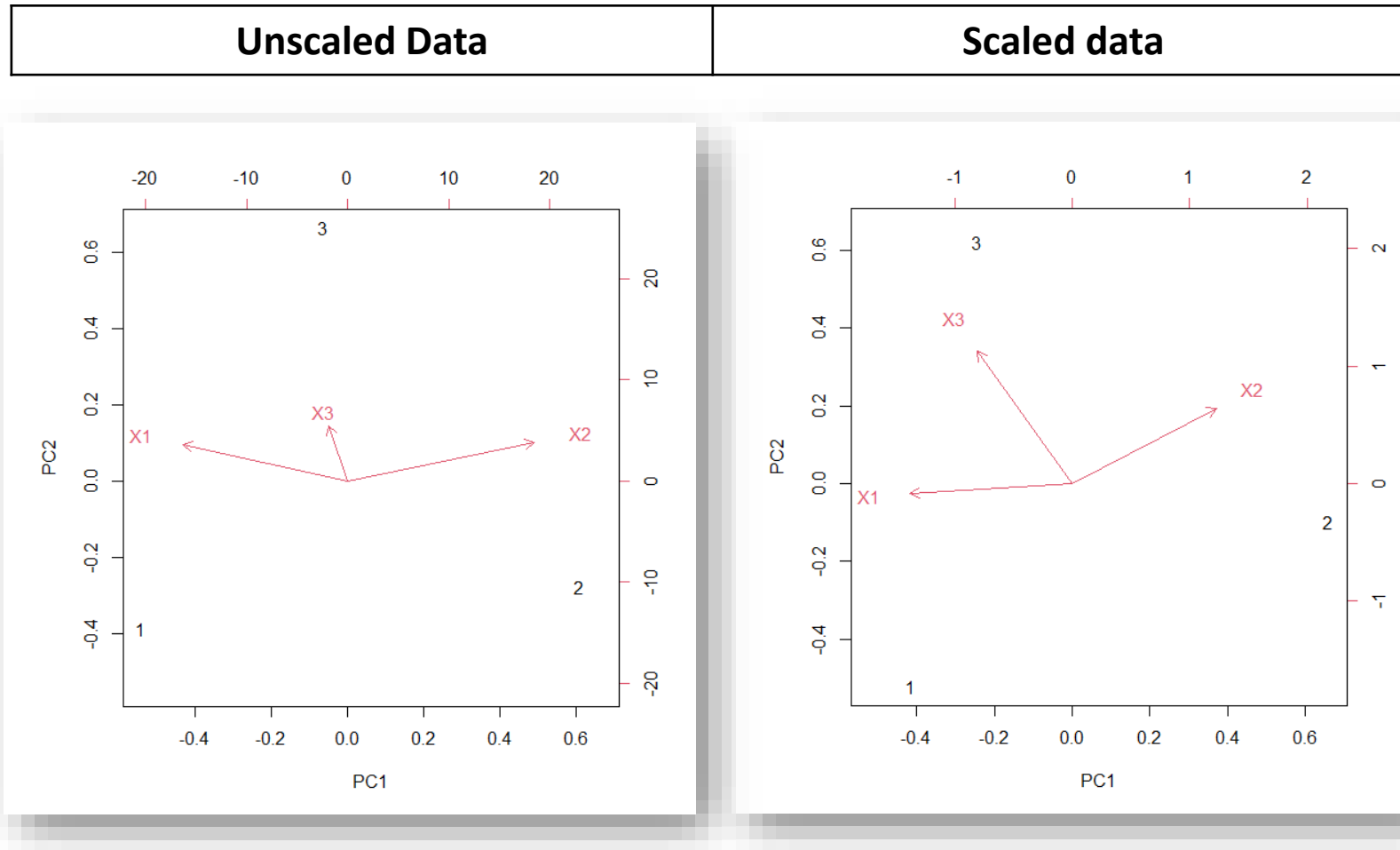


Principal Component Analysis: Contribution of Each Variable with Scaled Data

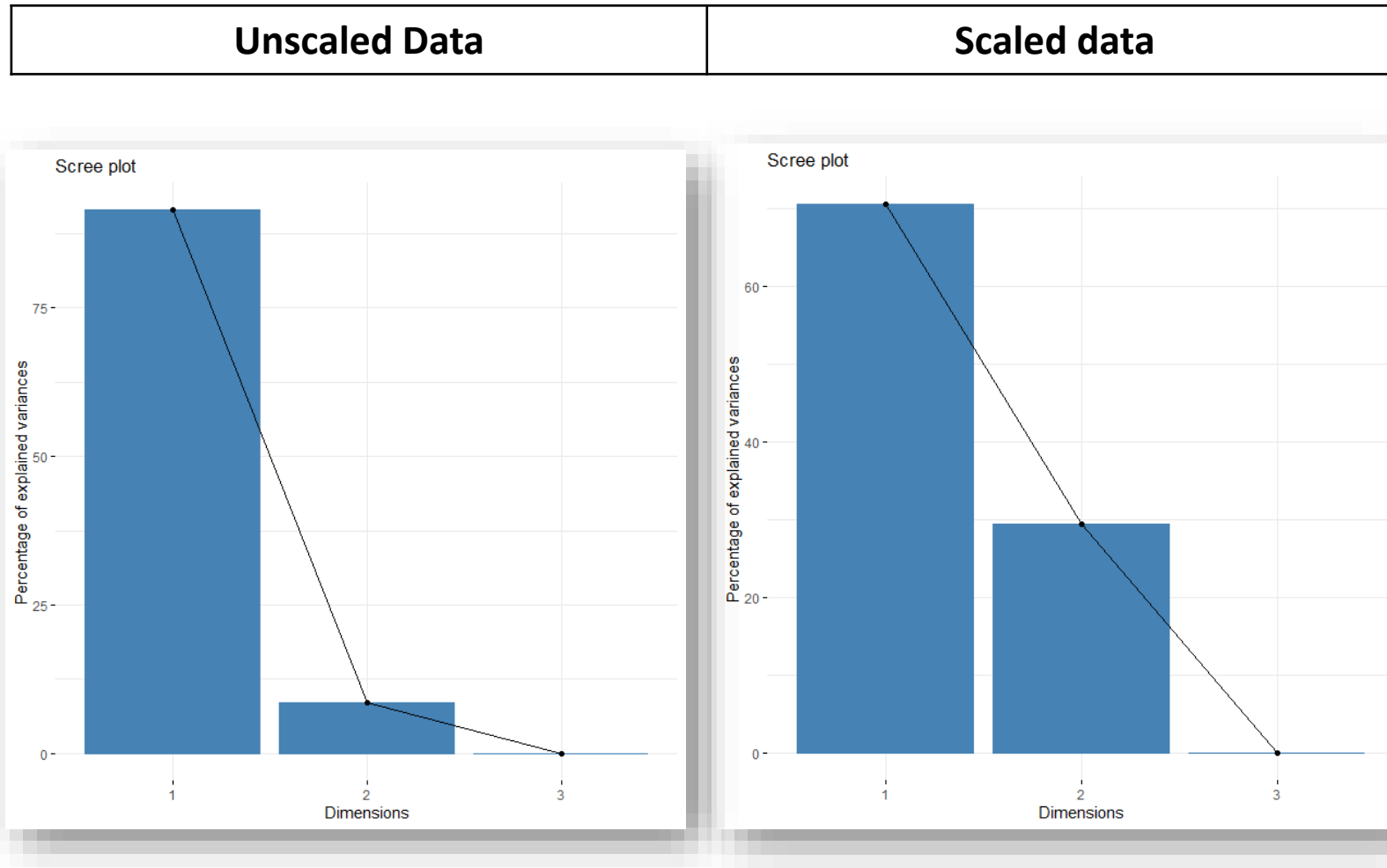
```
fviz_cos2(pcddataoutcs, choice = "var", axes = 1:2)
```



Principal Component Analysis : Example- Comparison of Biplots of Scaled and Unscaled Data



Principal Component Analysis : Example- Comparison of Scree Plots of Scaled and Unscaled Data



Principal Component Analysis: Example with USArrest Data

Use the built in data **USArrests** in R software

Violent Crime Rates by US State

Description : This data set contains statistics, in arrests per 100,000 residents for assault, murder, and rape in each of the 50 US states in 1973. Also given is the percent of the population living in urban areas.

Usage: **USArrests**

Format: A data frame with 50 observations on 4 variables.

[,1]	Murder	numeric	Murder arrests (per 100,000)
[,2]	Assault	numeric	Assault arrests (per 100,000)
[,3]	UrbanPop	numeric	Percent urban population
[,4]	Rape	numeric	Rape arrests (per 100,000)

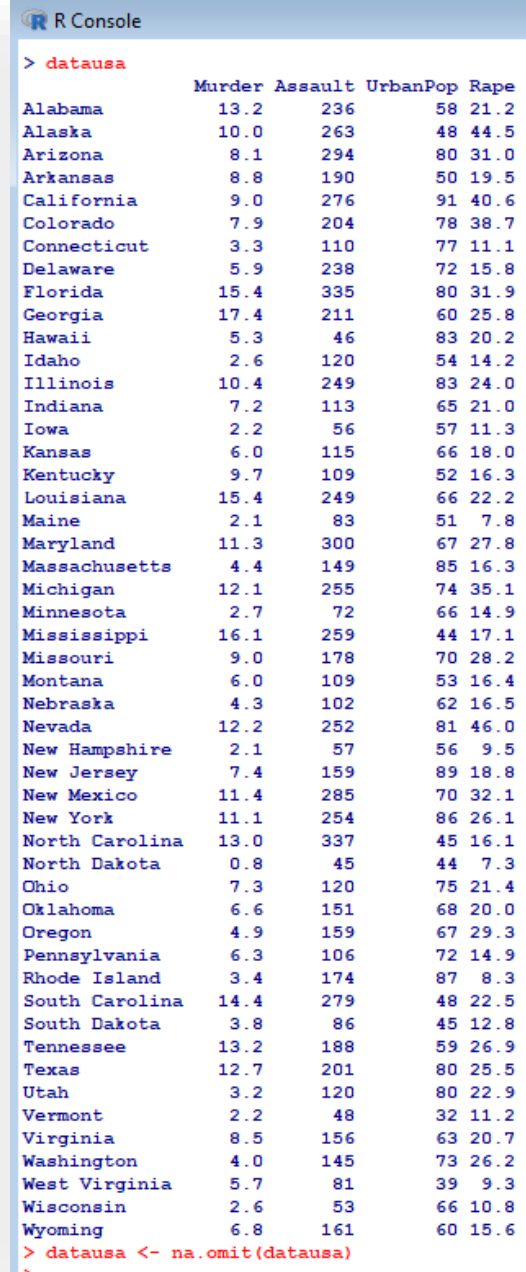
Note: **USArrests** contains the data as in McNeil's monograph. For the UrbanPop percentages, a review of the table (No. 21) in the Statistical Abstracts 1975 reveals a transcription error for Maryland (and that McNeil used the same "round to even" rule that R's round() uses), as found by Daniel S Coven (Arizona).

Source: World Almanac and Book of facts 1975. (Crime rates).

References: McNeil, D. R. (1977) Interactive Data Analysis. New York: Wiley.

Principal Component Analysis: Example with USArrest Data

Use the built in data `USArrests`.



The image shows a screenshot of an R console window. The title bar says "R Console". The prompt is "> datausa". Below it, a table of data is displayed. The table has 5 columns: "Murder", "Assault", "UrbanPop", and "Rape". The rows represent 50 US states, ordered alphabetically by state name. The data values are numerical. At the bottom of the console, the command "> datausa <- na.omit(datausa)" is entered, followed by a backslash character.

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7
Connecticut	3.3	110	77	11.1
Delaware	5.9	238	72	15.8
Florida	15.4	335	80	31.9
Georgia	17.4	211	60	25.8
Hawaii	5.3	46	83	20.2
Idaho	2.6	120	54	14.2
Illinois	10.4	249	83	24.0
Indiana	7.2	113	65	21.0
Iowa	2.2	56	57	11.3
Kansas	6.0	115	66	18.0
Kentucky	9.7	109	52	16.3
Louisiana	15.4	249	66	22.2
Maine	2.1	83	51	7.8
Maryland	11.3	300	67	27.8
Massachusetts	4.4	149	85	16.3
Michigan	12.1	255	74	35.1
Minnesota	2.7	72	66	14.9
Mississippi	16.1	259	44	17.1
Missouri	9.0	178	70	28.2
Montana	6.0	109	53	16.4
Nebraska	4.3	102	62	16.5
Nevada	12.2	252	81	46.0
New Hampshire	2.1	57	56	9.5
New Jersey	7.4	159	89	18.8
New Mexico	11.4	285	70	32.1
New York	11.1	254	86	26.1
North Carolina	13.0	337	45	16.1
North Dakota	0.8	45	44	7.3
Ohio	7.3	120	75	21.4
Oklahoma	6.6	151	68	20.0
Oregon	4.9	159	67	29.3
Pennsylvania	6.3	106	72	14.9
Rhode Island	3.4	174	87	8.3
South Carolina	14.4	279	48	22.5
South Dakota	3.8	86	45	12.8
Tennessee	13.2	188	59	26.9
Texas	12.7	201	80	25.5
Utah	3.2	120	80	22.9
Vermont	2.2	48	32	11.2
Virginia	8.5	156	63	20.7
Washington	4.0	145	73	26.2
West Virginia	5.7	81	39	9.3
Wisconsin	2.6	53	66	10.8
Wyoming	6.8	161	60	15.6

```
> datausa <- na.omit(datausa)
\
```

Principal Component Analysis: Example with USArrest Data

```
data("USArrests") #load data
```

```
head(USArrests) #view first six rows of data
```

	Murder	Assault	UrbanPop	Rape
Alabama	13.2	236	58	21.2
Alaska	10.0	263	48	44.5
Arizona	8.1	294	80	31.0
Arkansas	8.8	190	50	19.5
California	9.0	276	91	40.6
Colorado	7.9	204	78	38.7

Principal Component Analysis: Example with USArrest Data

```
pcdataUSarrest = prcomp(USArrests)
```

```
> pcdataUSarrest
```

```
Standard deviations (1, ..., p=4):
```

```
[1] 83.732400 14.212402  6.489426  2.482790
```

```
Rotation (n x k) = (4 x 4):
```

	PC1	PC2	PC3	PC4
Murder	0.04170432	-0.04482166	0.07989066	-0.99492173
Assault	0.99522128	-0.05876003	-0.06756974	0.03893830
UrbanPop	0.04633575	0.97685748	-0.20054629	-0.05816914
Rape	0.07515550	0.20071807	0.97408059	0.07232502

The principal components are

$$Y_1 = 0.04 \text{ Murder} + 0.99 \text{ Assault} + 0.05 \text{ UrbanPop} + 0.08 \text{ Rape}$$

$$Y_2 = -0.04 \text{ Murder} - 0.05 \text{ Assault} + 0.97 \text{ UrbanPop} + 0.20 \text{ Rape}$$

$$Y_3 = 0.08 \text{ Murder} - 0.07 \text{ Assault} - 0.20 \text{ UrbanPop} + 0.97 \text{ Rape}$$

$$Y_4 = -0.99 \text{ Murder} + 0.04 \text{ Assault} - 0.06 \text{ UrbanPop} + 0.07 \text{ Rape}$$

Principal Component Analysis: Example with USArrest Data

```
> summary(pcdDataUSArrest)
```

Importance of components:

	PC1	PC2	PC3	PC4
Standard deviation	83.7324	14.21240	6.4894	2.48279
Proportion of Variance	0.9655	0.02782	0.0058	0.00085
Cumulative Proportion	0.9655	0.99335	0.9991	1.00000

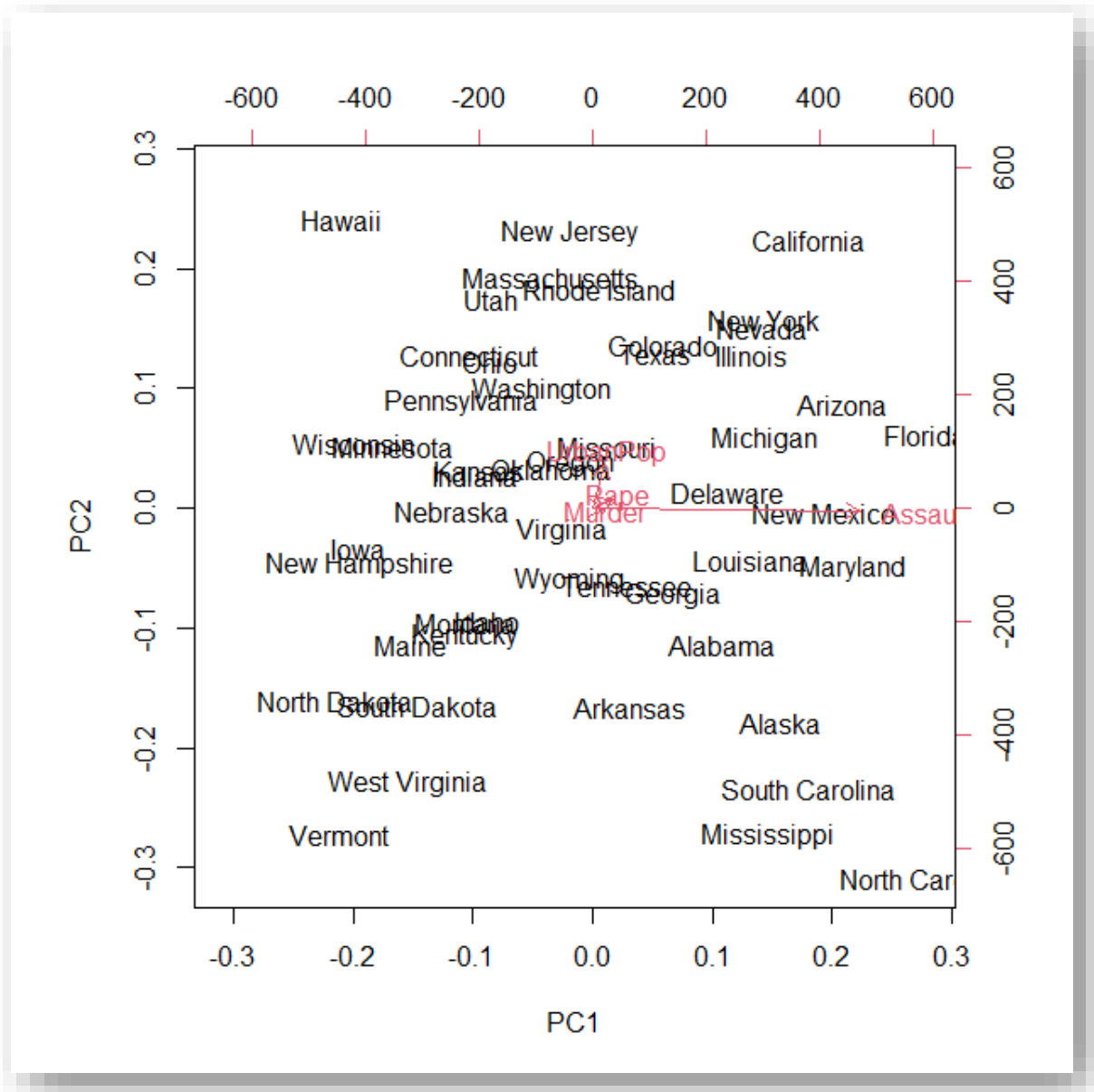
Principal Component Analysis: Example with USArrest Data

```
R Console
> data("USArrests") #load data
> head(USArrests) #view first six rows of data
      Murder Assault UrbanPop Rape
Alabama    13.2     236      58 21.2
Alaska     10.0     263      48 44.5
Arizona     8.1     294      80 31.0
Arkansas    8.8     190      50 19.5
California  9.0     276      91 40.6
Colorado   7.9     204      78 38.7
> pcdataUSarrest = prcomp(USArrests)
> pcdataUSarrest
Standard deviations (1, ..., p=4):
[1] 83.732400 14.212402  6.489426  2.482790

Rotation (n x k) = (4 x 4):
      PC1      PC2      PC3      PC4
Murder  0.04170432 -0.04482166  0.07989066 -0.99492173
Assault  0.99522128 -0.05876003 -0.06756974  0.03893830
UrbanPop 0.04633575  0.97685748 -0.20054629 -0.05816914
Rape     0.07515550  0.20071807  0.97408059  0.07232502
> summary(pcdataUSarrest)
Importance of components:
      PC1      PC2      PC3      PC4
Standard deviation    83.7324 14.21240 6.4894 2.48279
Proportion of Variance 0.9655  0.02782 0.0058 0.00085
Cumulative Proportion 0.9655  0.99335 0.9991 1.00000
> |
```

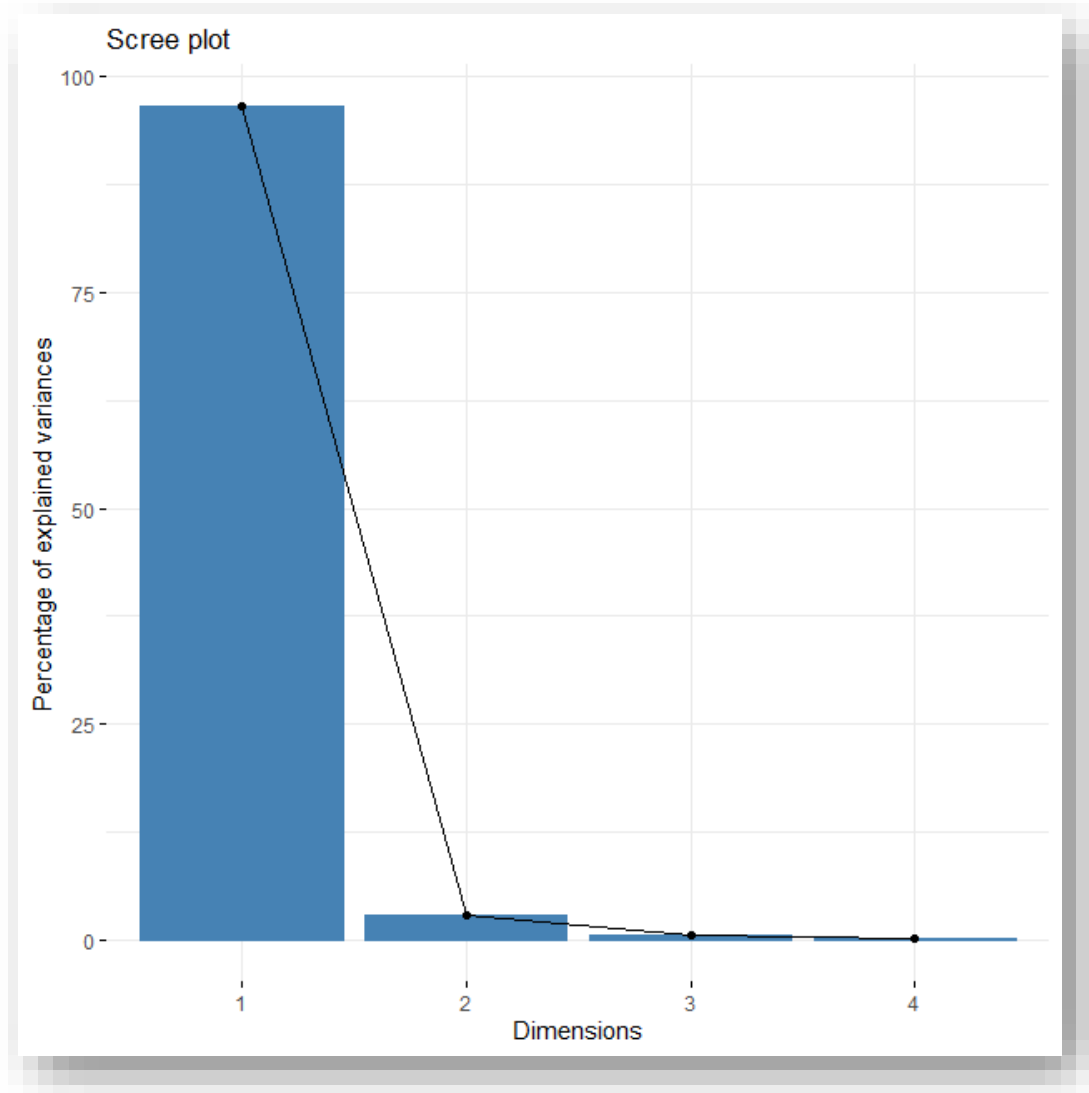
Principal Component Analysis: Biplot with USArrest Data

```
> biplot(pcdDataUSArrest)
```



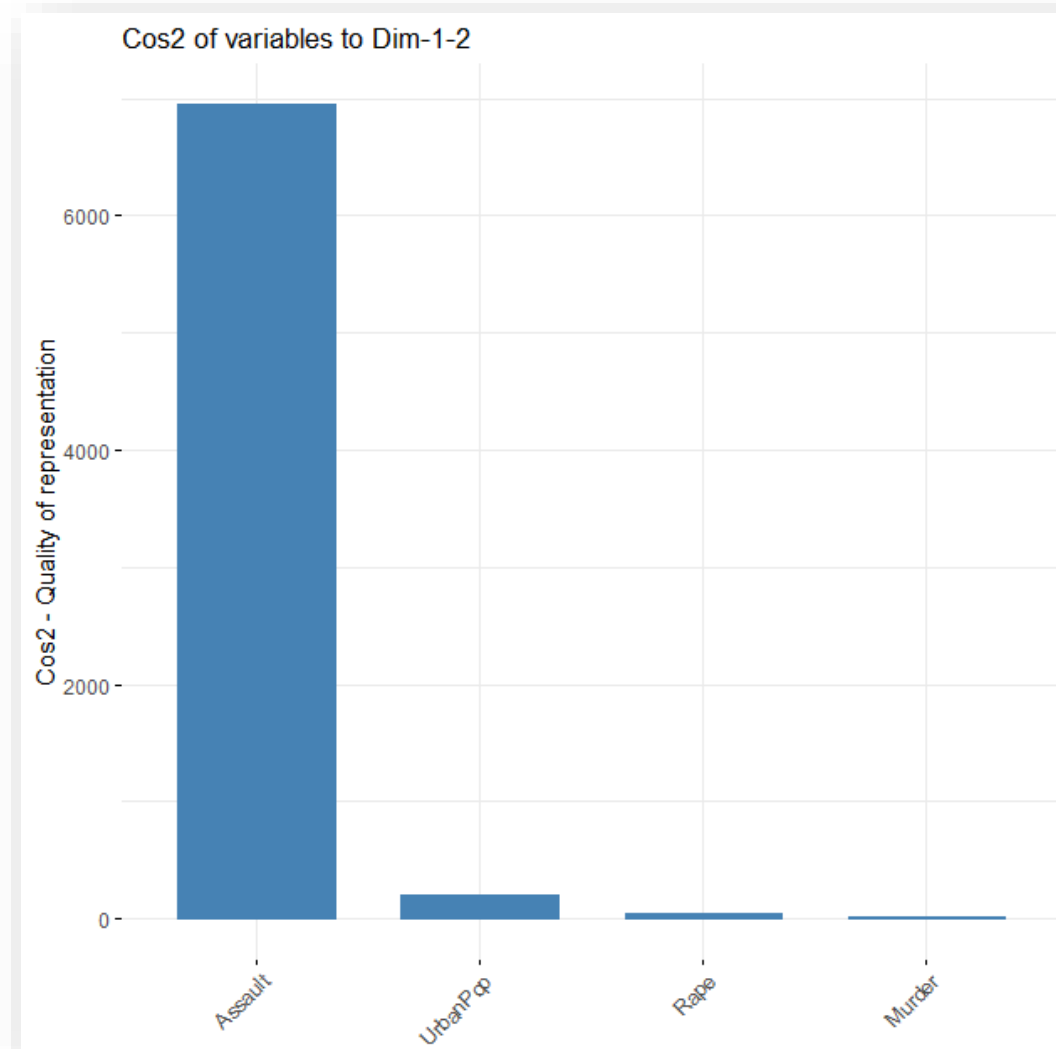
Principal Component Analysis: Scree Plot with USArrest Data

`fviz_eig(pcdatUSarrest)`



Principal Component Analysis: Cos Plot with USArrest Data

```
fviz_cos2(pcddataUSarrest, choice = "var", axes = 1:2)
```



Principal Component Analysis: Scree and Cos Plots with USArrest Data

```
fviz_pca_var(pcddataUSarrest, col.var = "cos2",  
gradient.cols = c("blue", "red", "green",  
"magenta"), repel = TRUE)
```

